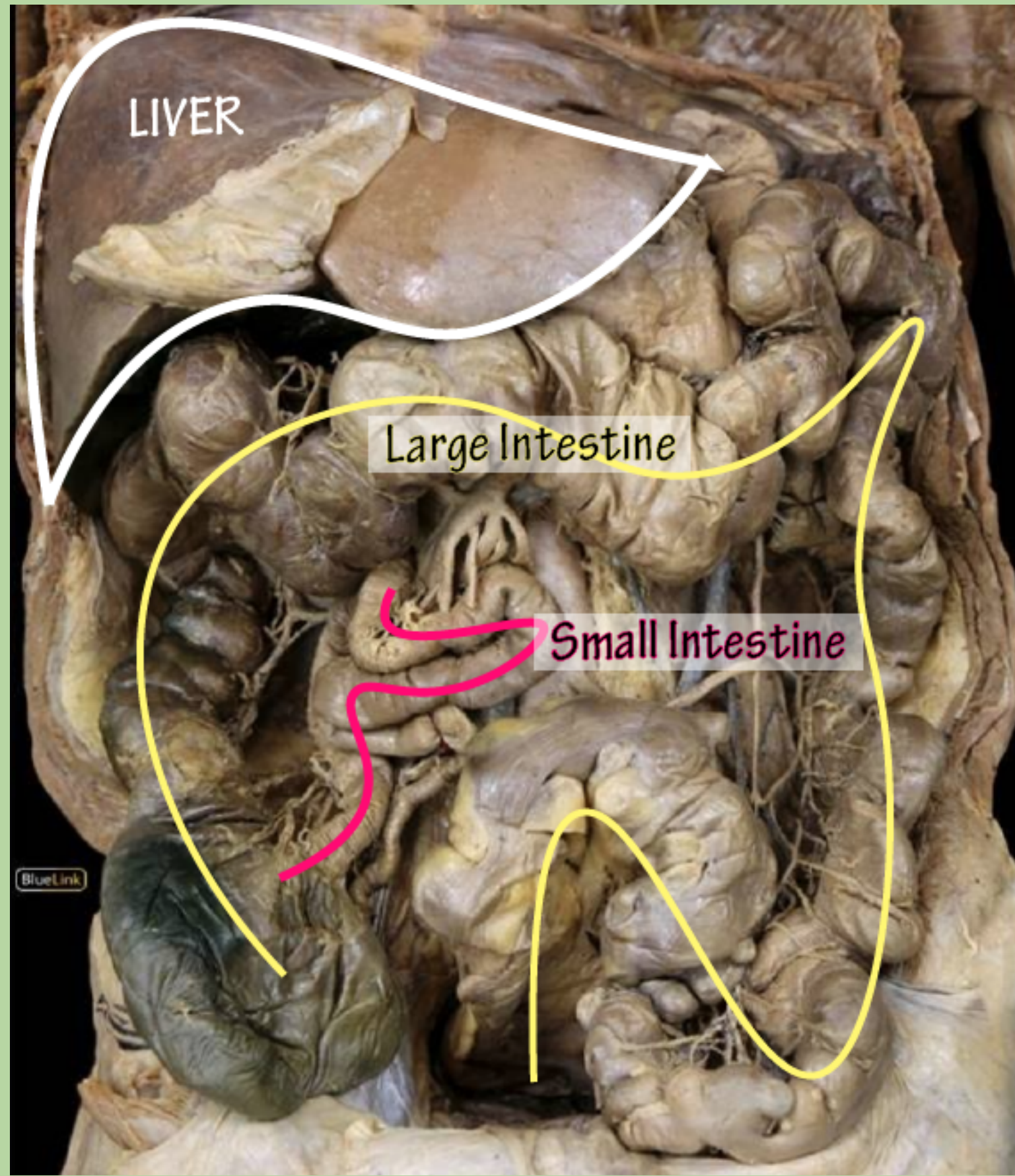
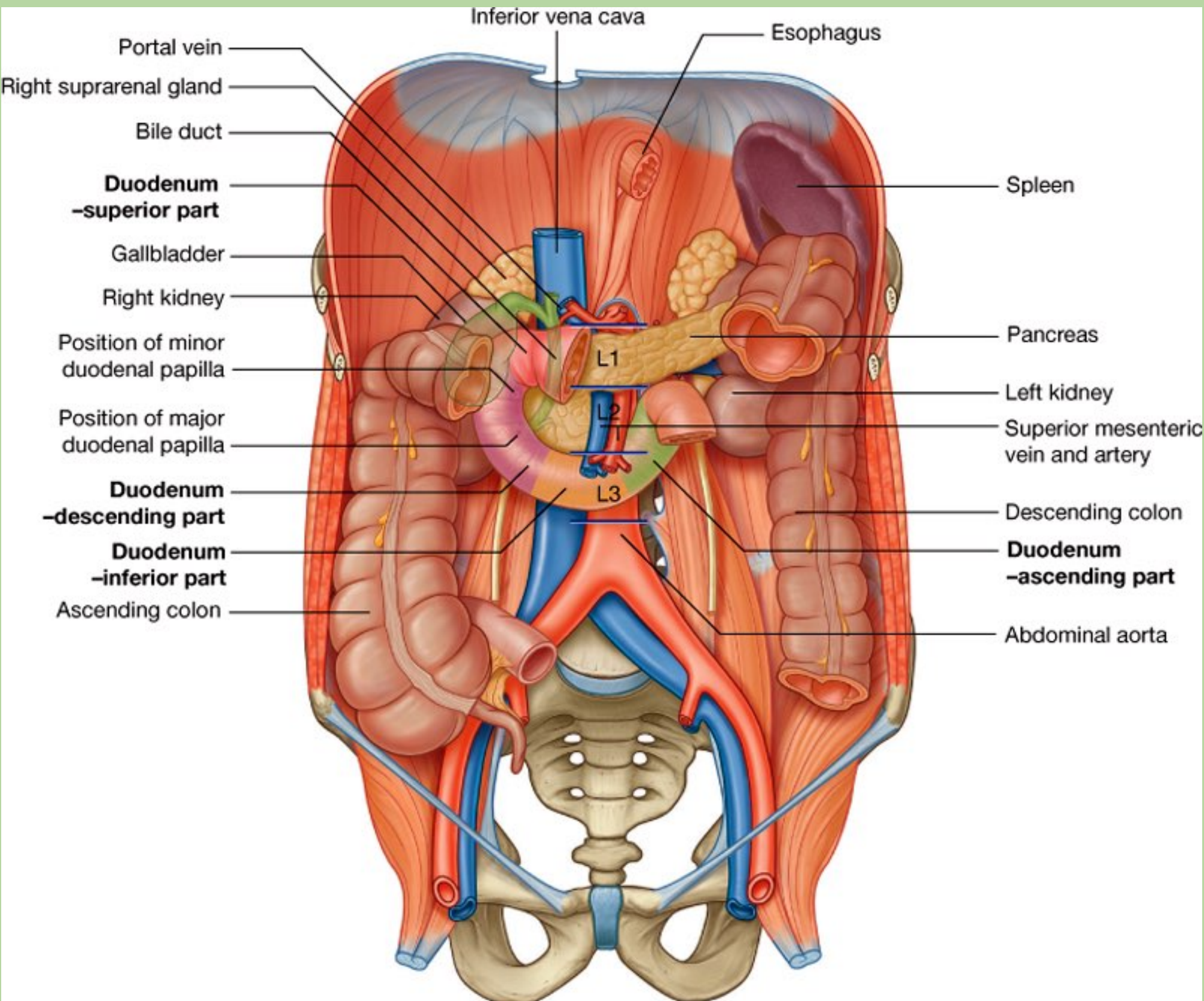


Small intestine

- approx. length: 6 m
- **Parts:**
duodenum,
jejunum and ileum



Duodenum

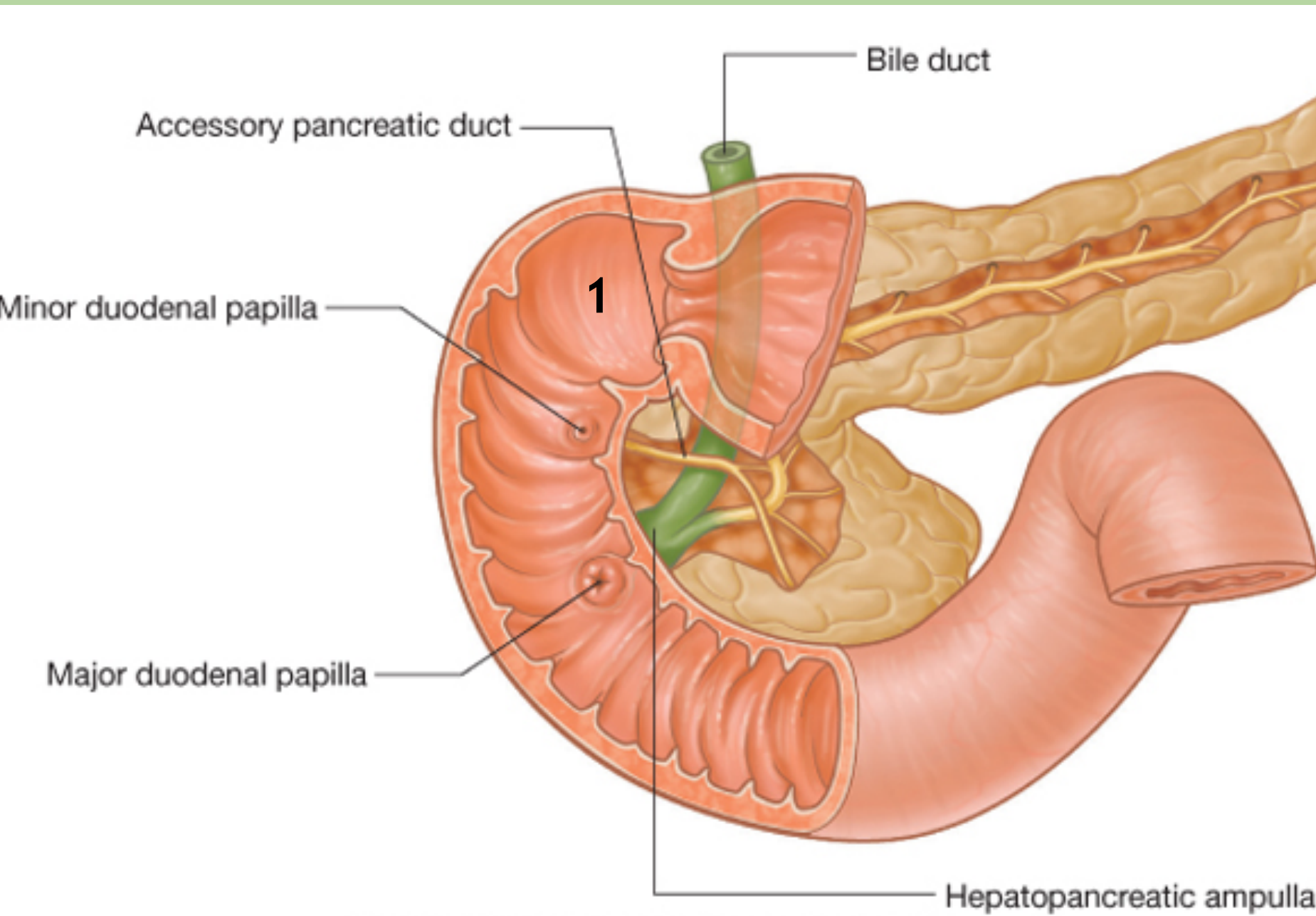


- approx. length: 25-30 cm
- vertebral levels: L1-3
- C-shaped curve around the head of the pancreas

• Parts:

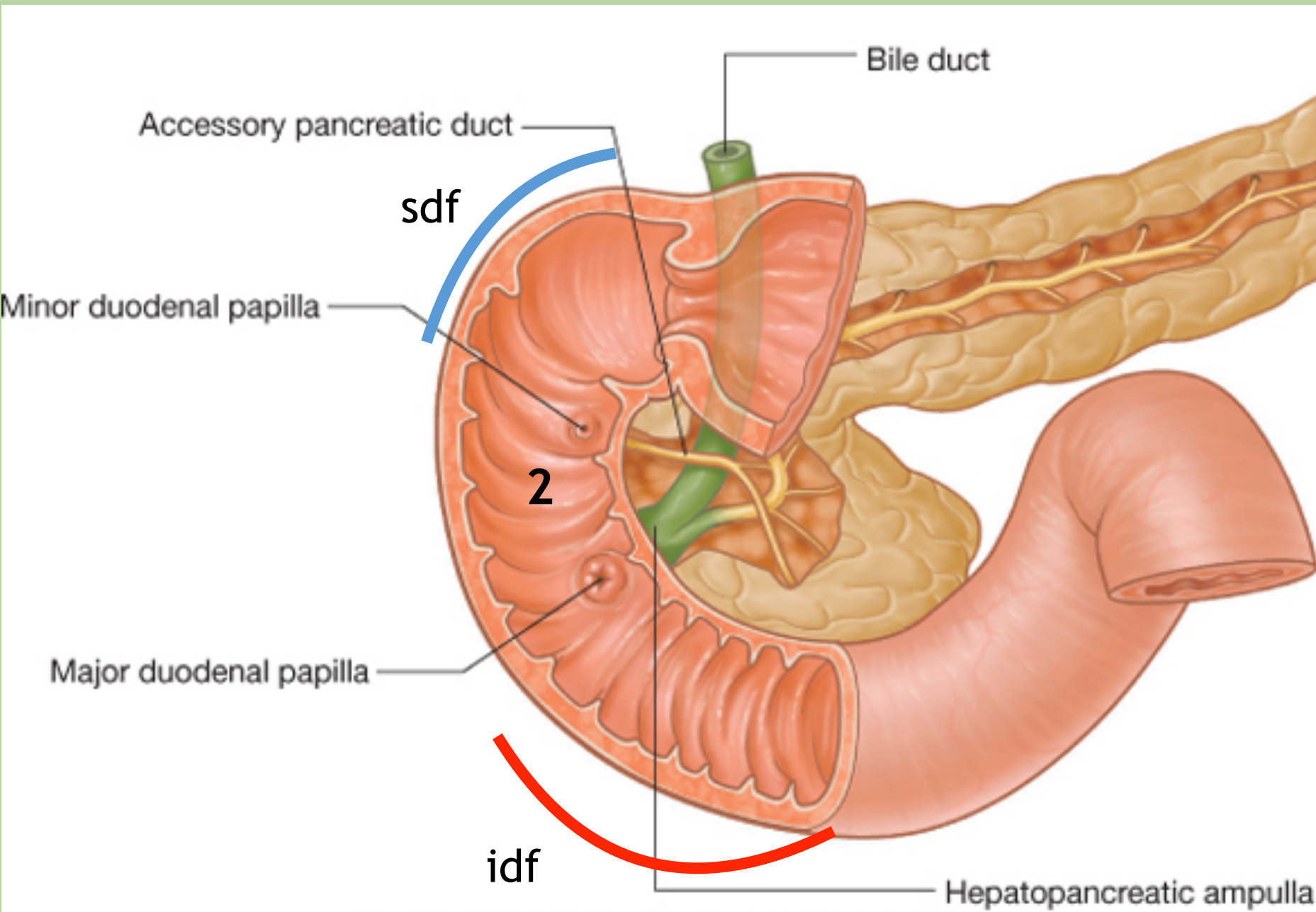
- 1: superior part
- 2: descending part
- 3: inferior (horizontal) part
- 4: ascending part

Duodenum: superior part (1)



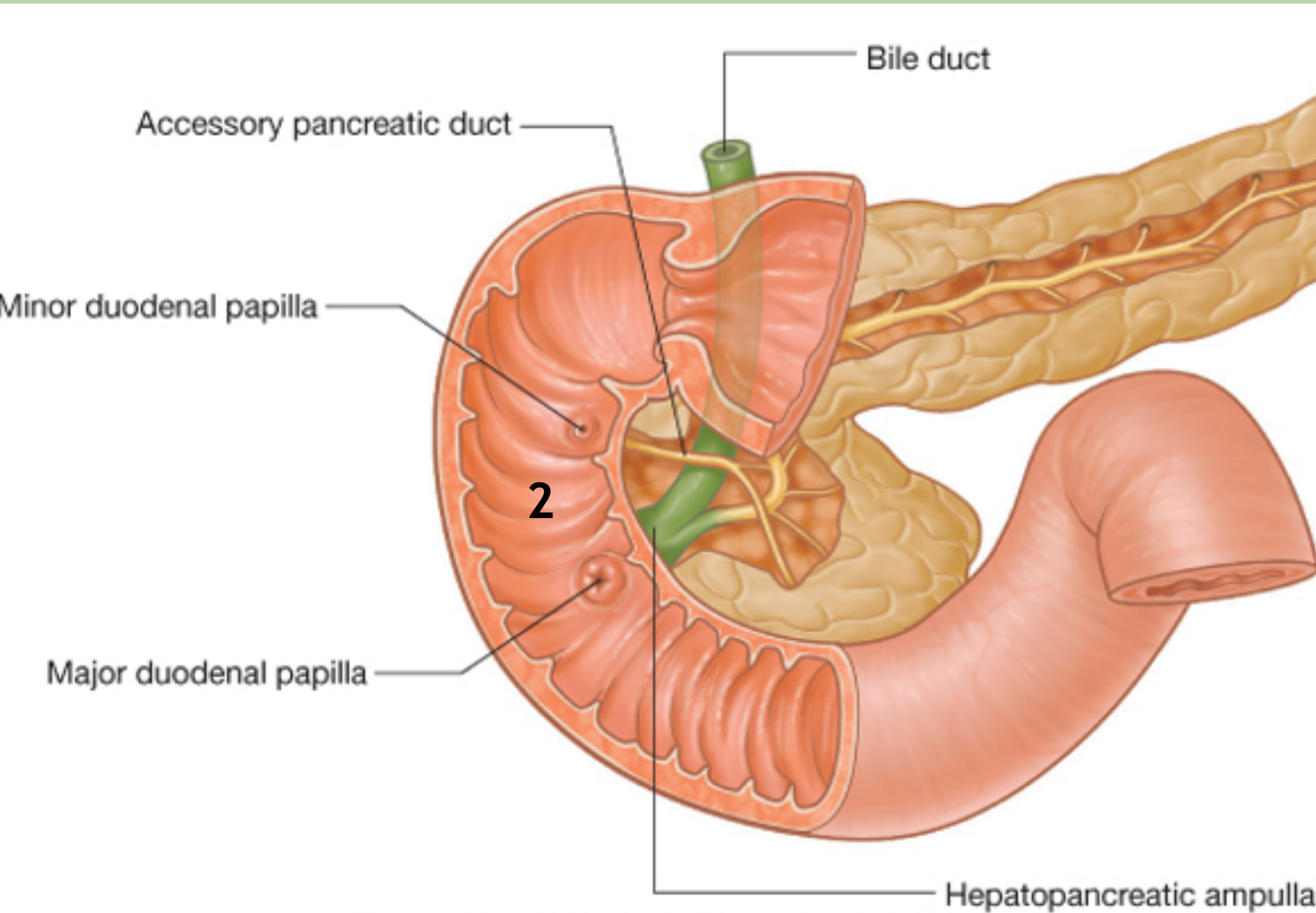
- L1 level
- *hepatoduodenal ligament attaches to its upper surface*
- **intraperitoneal structure**
- wide proximal lumen: *ampulla (=duodenal bulb)*

Duodenum: descending part (2)



- under the superior duodenal flexure (sdf) to the inferior duodenal flexure (idf)
- L1-3 levels
- **s e c o n d a r i l y**
r e t r o p e r i t o n e a l
s t r u c t u r e
- c i r c u l a r a n d
longitudinal mucosal folds
- *minor duodenal papilla*
- *major duodenal papilla*
(=Vater's papilla)

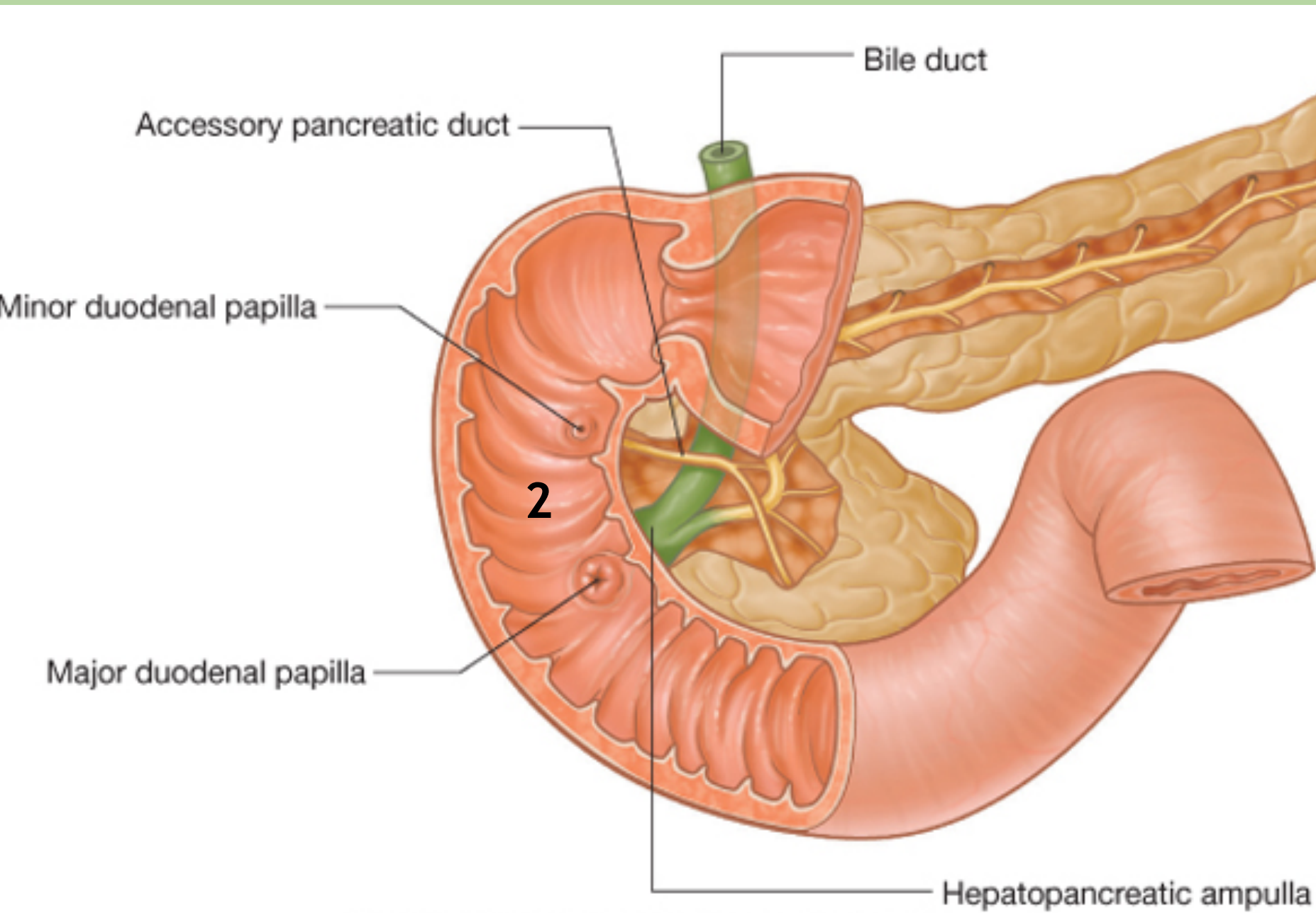
Duodenum: descending part (2)



Major duodenal papilla (=Vater's papilla):

- it locates 10 cm inferior to the pylorus
- the *hepatopancreatic ampulla*, formed by the *common bile duct and the major pancreatic duct* (=Wirsung's duct), enters the duodenum through the Vater's papilla on the postero-medial wall
- Oddi sphincter muscle lets the juices (bile and pancreatic juice) enter the duodenum

Duodenum: descending part (2)



- the accessory pancreatic duct (Santorini's duct) enters through the **minor duodenal papilla**
- it is presented in the 70% of the population (*in the 30%, the accessory pancreatic duct empties into the major pancreatic duct*)

Accessory pancreatic duct



Hepatopancreatic
ampulla, deep

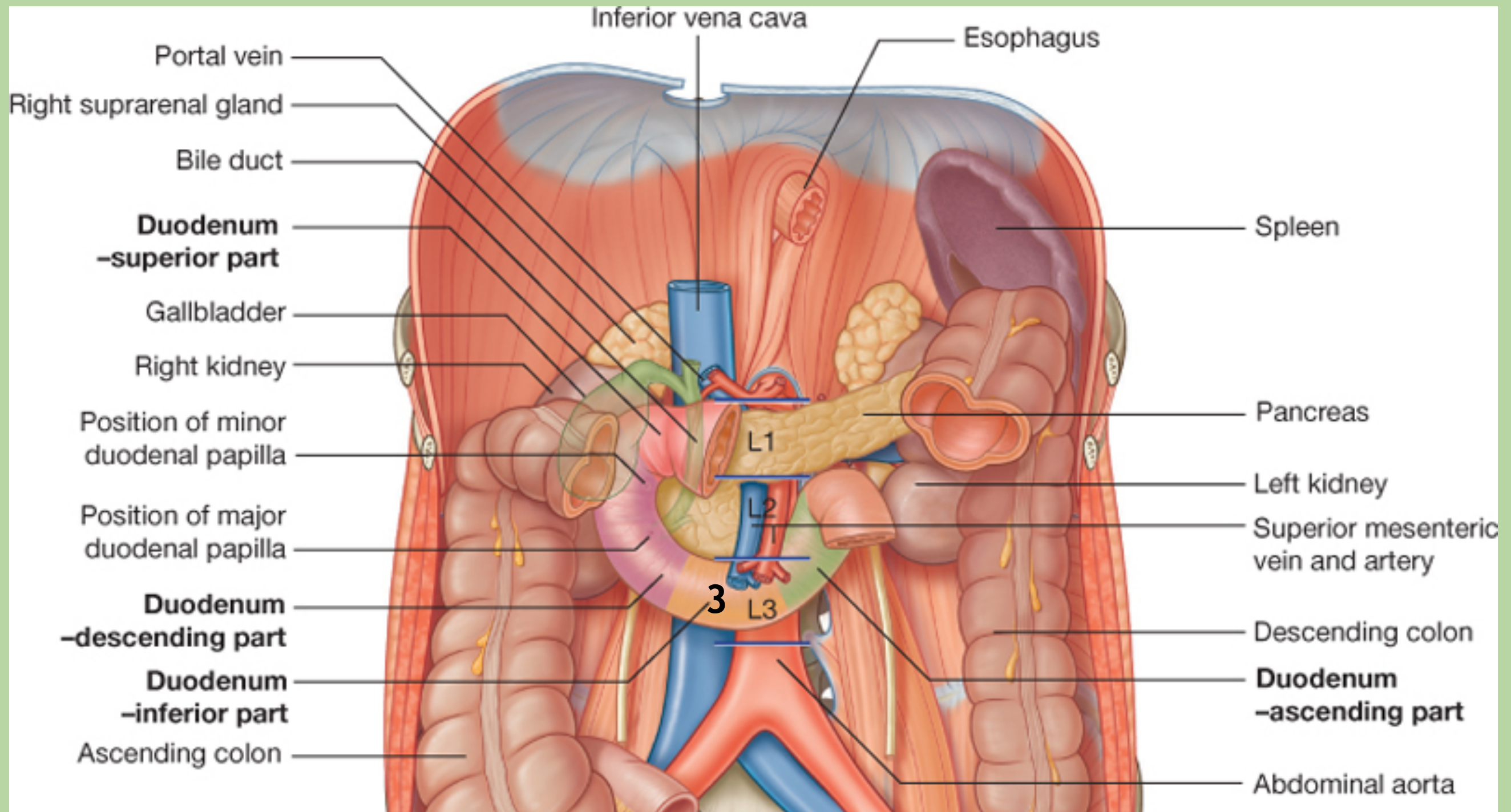
Major pancreatic duct

Abdominal cavity, anterior

BlueLink

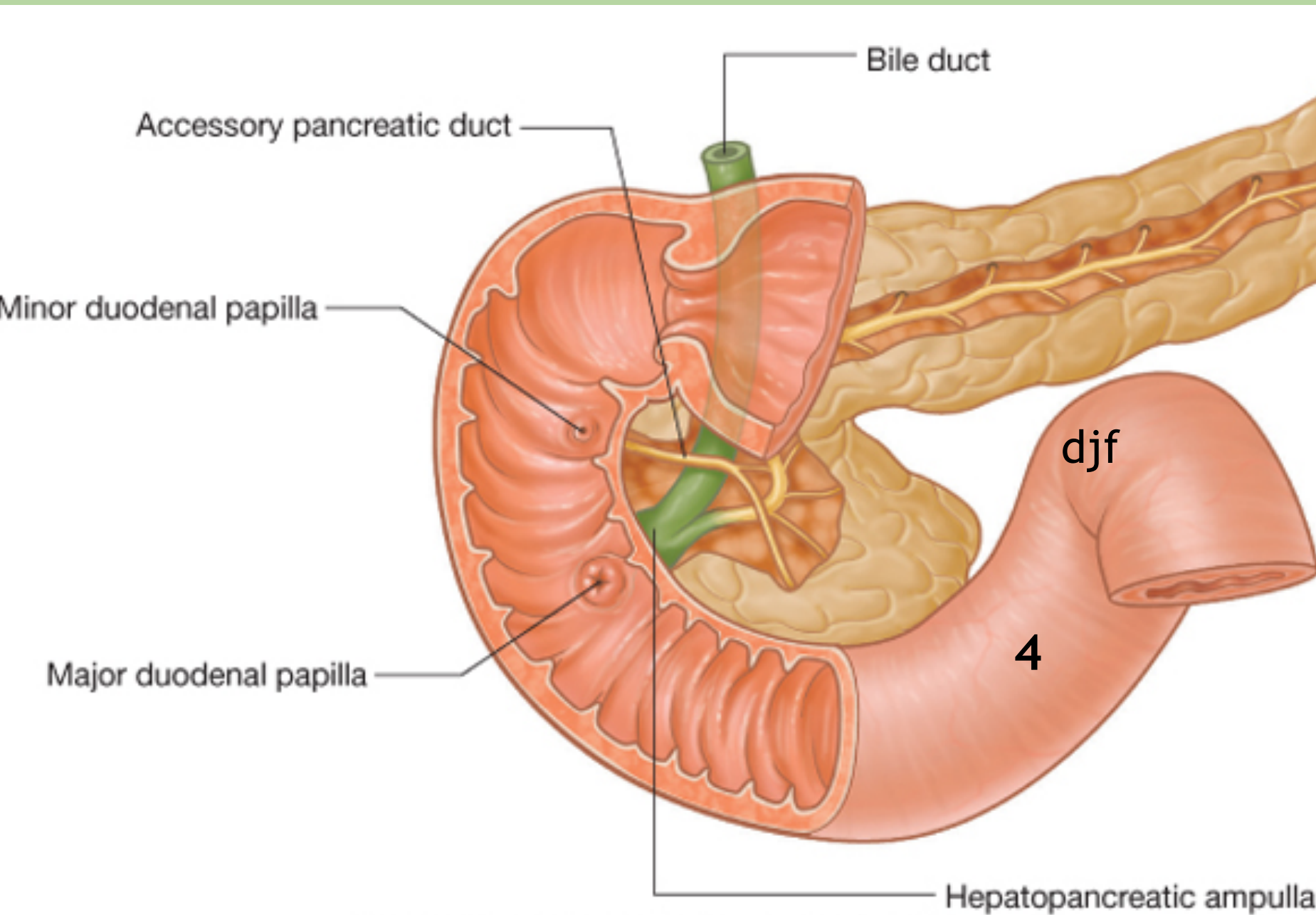
© B. Kathleen Alsup & Glenn M. Fox

Duodenum: inferior (horizontal) part (3)



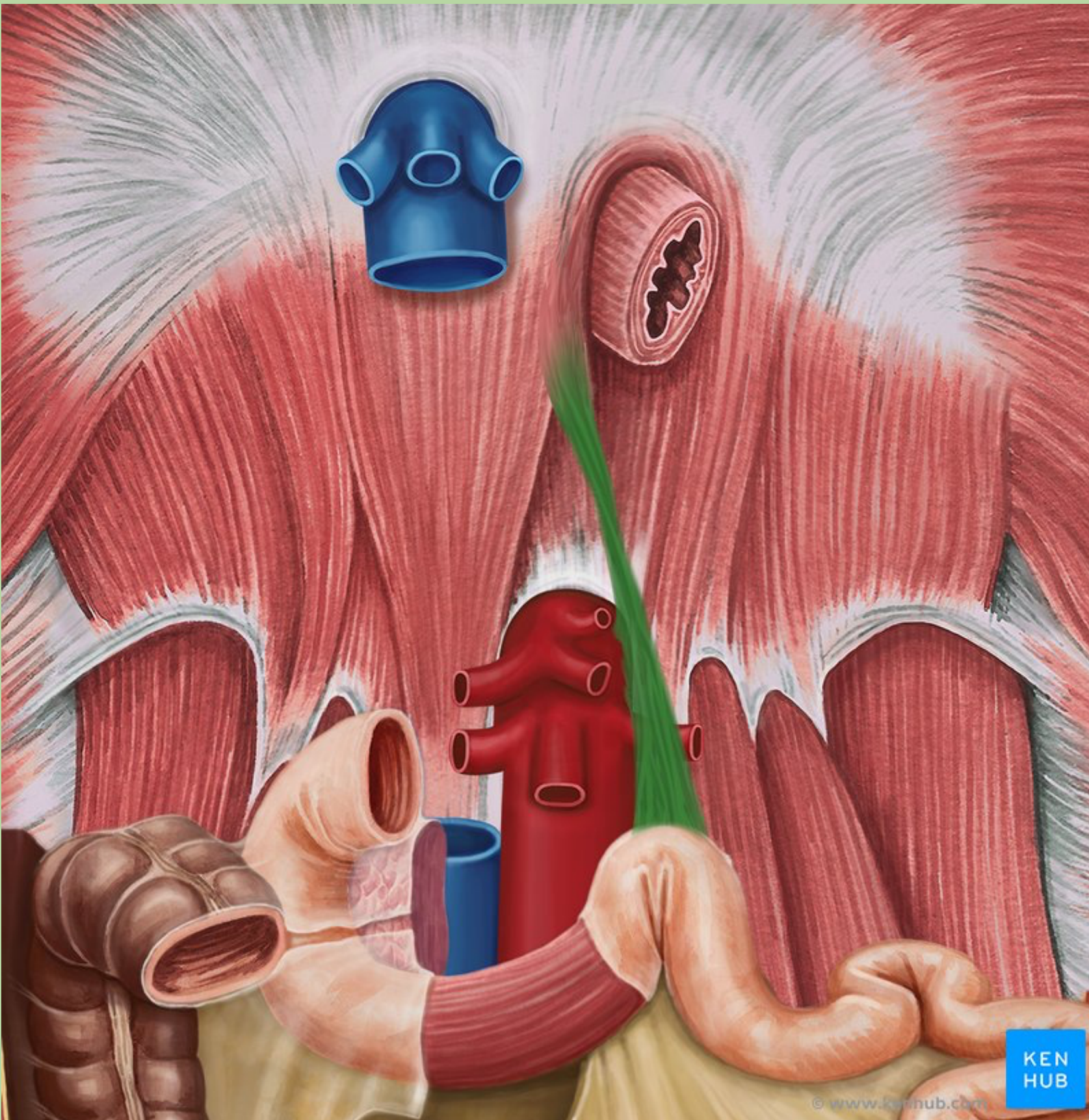
- L3 level
- **secondarily retroperitoneal structure**
- circular mucosal folds
- location: in front of the inferior vena cava and aorta; behind the superior mesenteric vessels

Duodenum: ascending part (4)



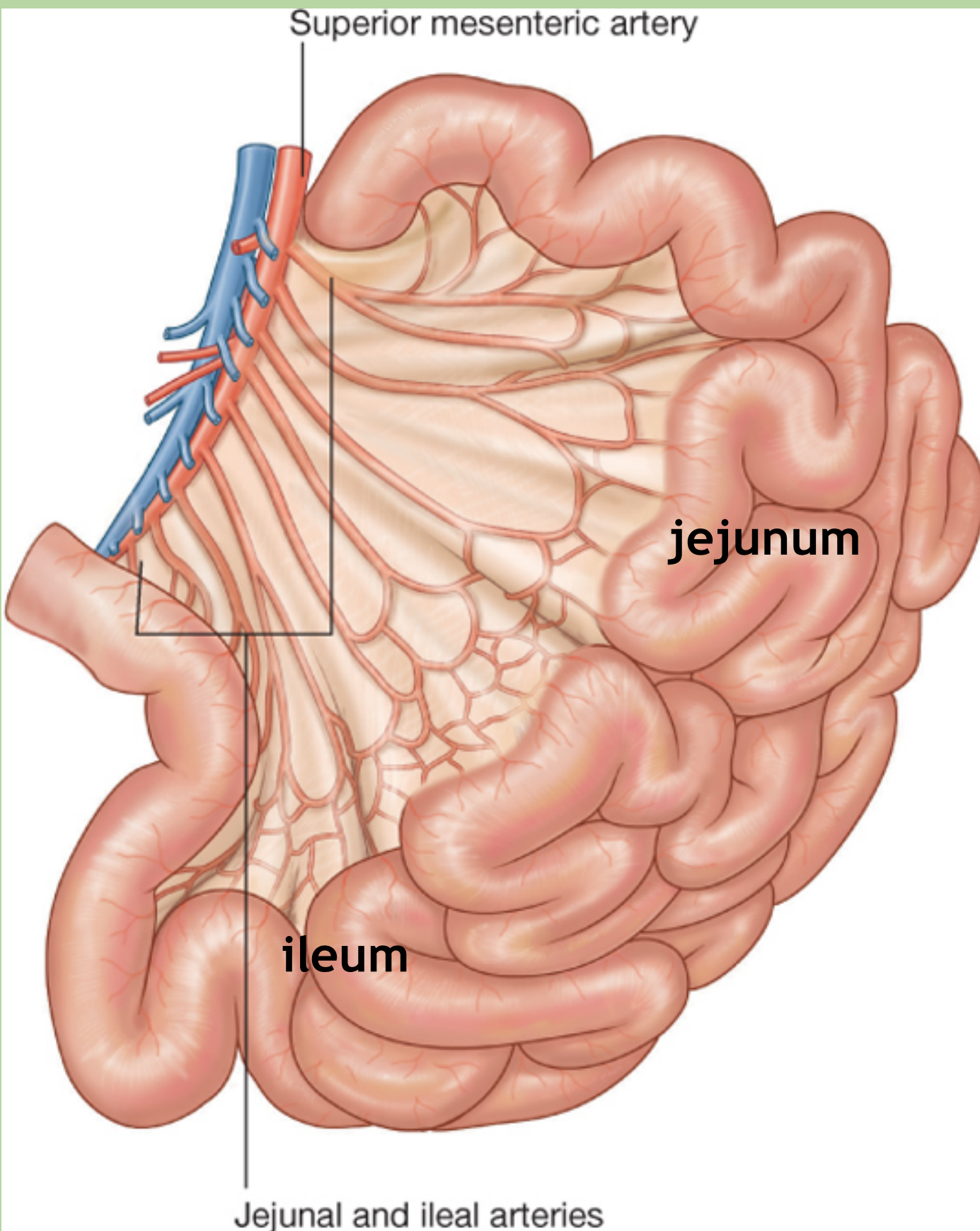
- L2 level
- **secondarily retroperitoneal**
- circular mucosal folds
- ends at the *duodenojejunal flexure (djf)*

Treitz's ligament (suspensory ligament of duodenum)



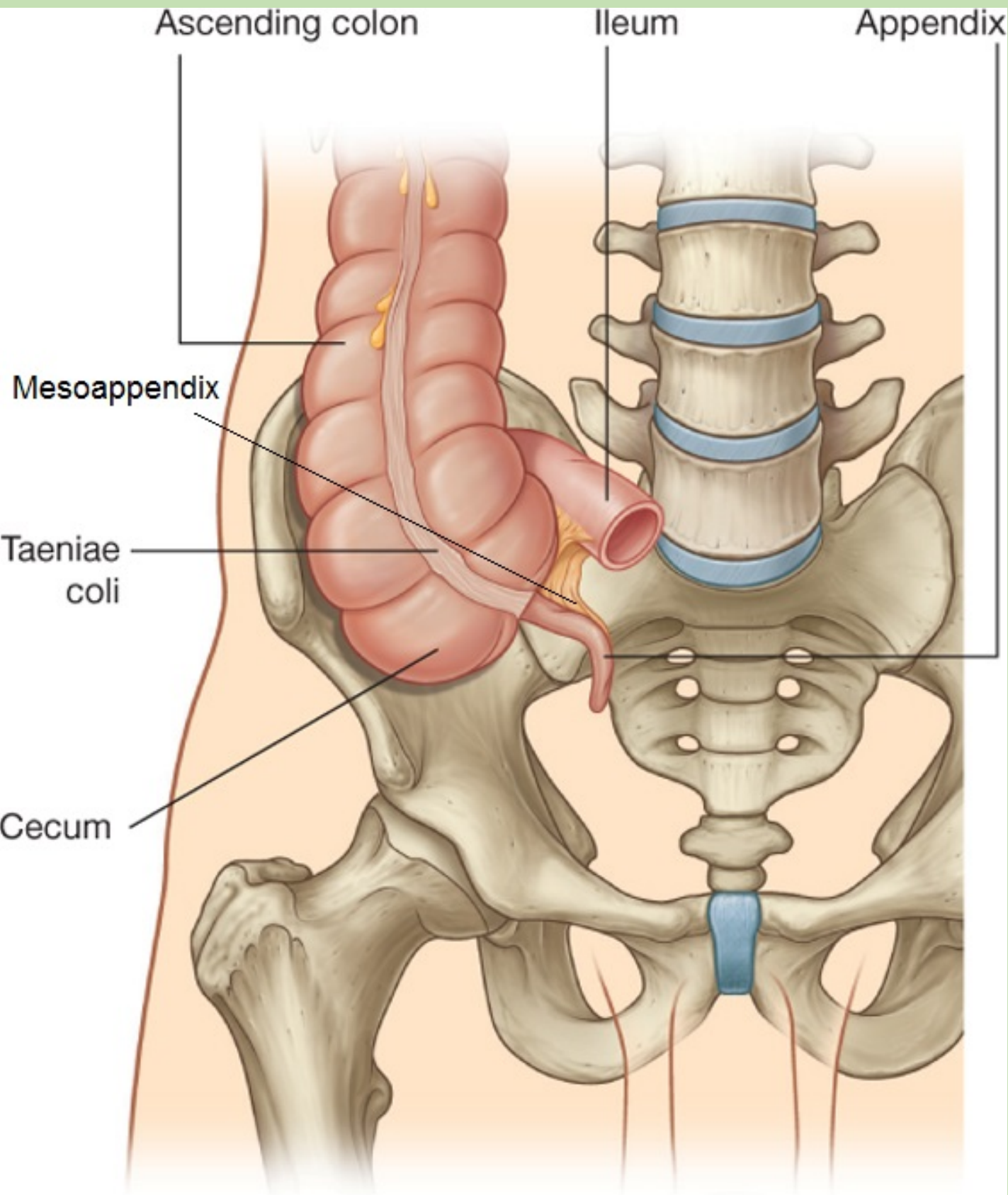
- peritoneal ligament which spans between the right diaphragmatic crus and duodenojejunal flexure
- *important landmark for discriminating between upper and lower gastrointestinal bleeding*
- *Treitz's herniation: small intestinal loops can get behind the ligament, causing obstruction or strangulation of the intestinal vessels*

Jejunum and ileum

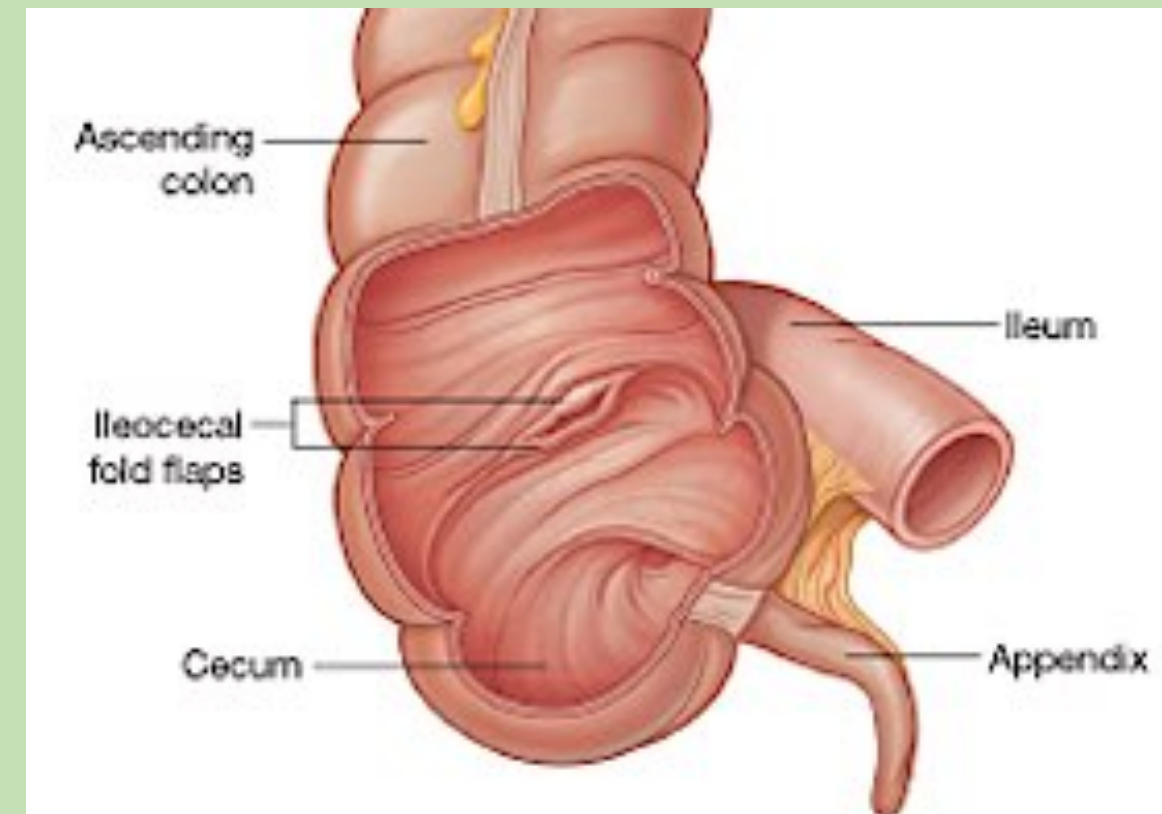


- **intraperitoneal** structures
- their total length: 3-5 m
- their form *intestinal loops*
- only the terminal ileum has relatively constant position
- *there is no macroscopic landmark between the two parts*
- *circular mucosal folds internally*

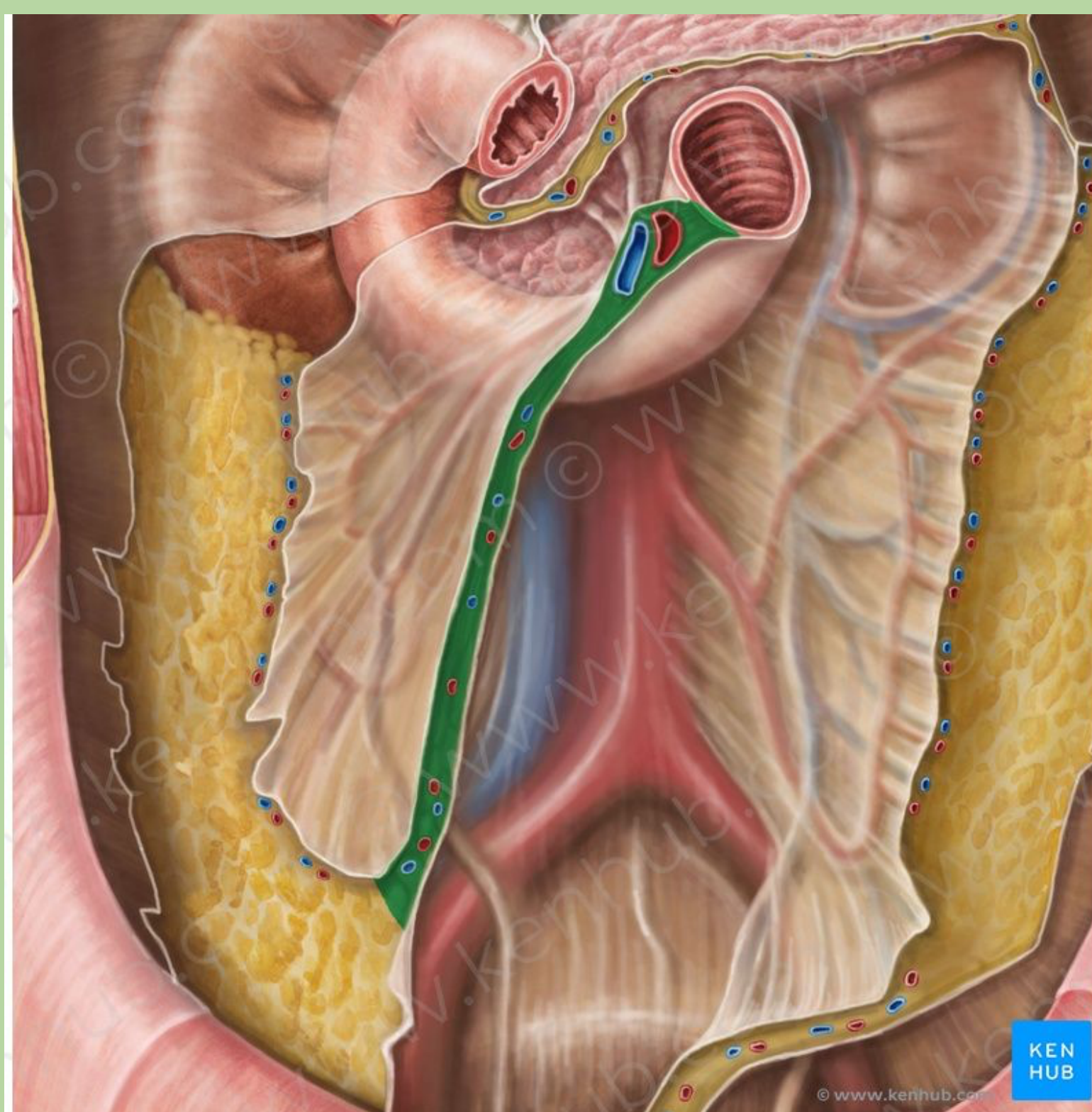
Ileocecal valve (Bauhin's valve)



The partly digested food, the chyme, flows into the cecum through the ileocecal valve. It prevents the reflux from the cecum.



Radix mesenterii (root of mesentery)



Mesentery: peritoneal duplication suspending the jejunum and ileum.

Root of mesentery: The attachment of the mesentery *to the posterior abdominal wall, extending from the duodenojejunal flexure to the ileocecal junction.*

Neurovasculature of the duodenum

Arterial supply:

- *superior anterior pancreaticoduodenal artery*
- *superior posterior pancreaticoduodenal artery*
- *inferior pancreaticoduodenal artery*

Venous drainage:

into the hepatic portal vein

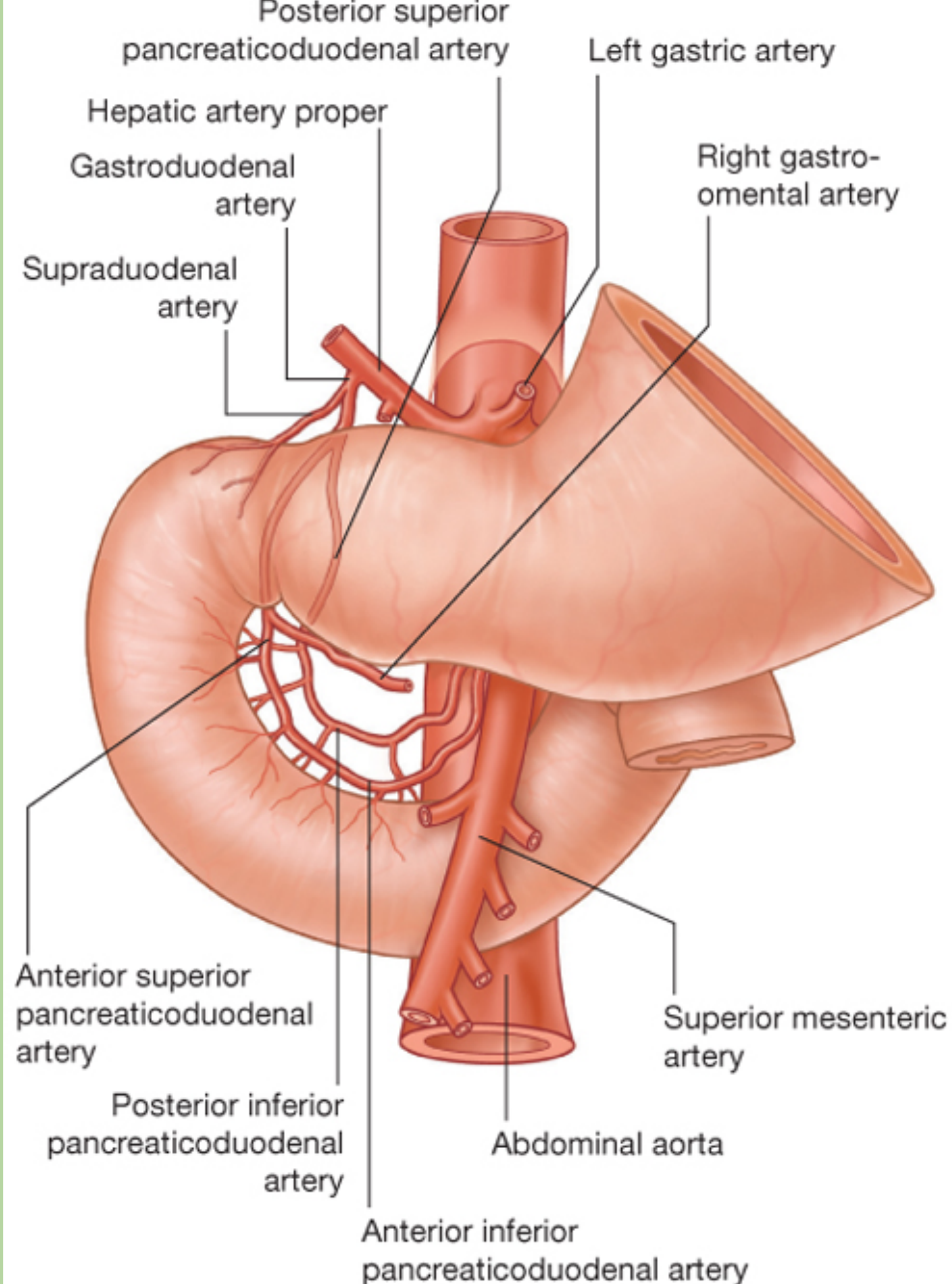
Lymphatic drainage:

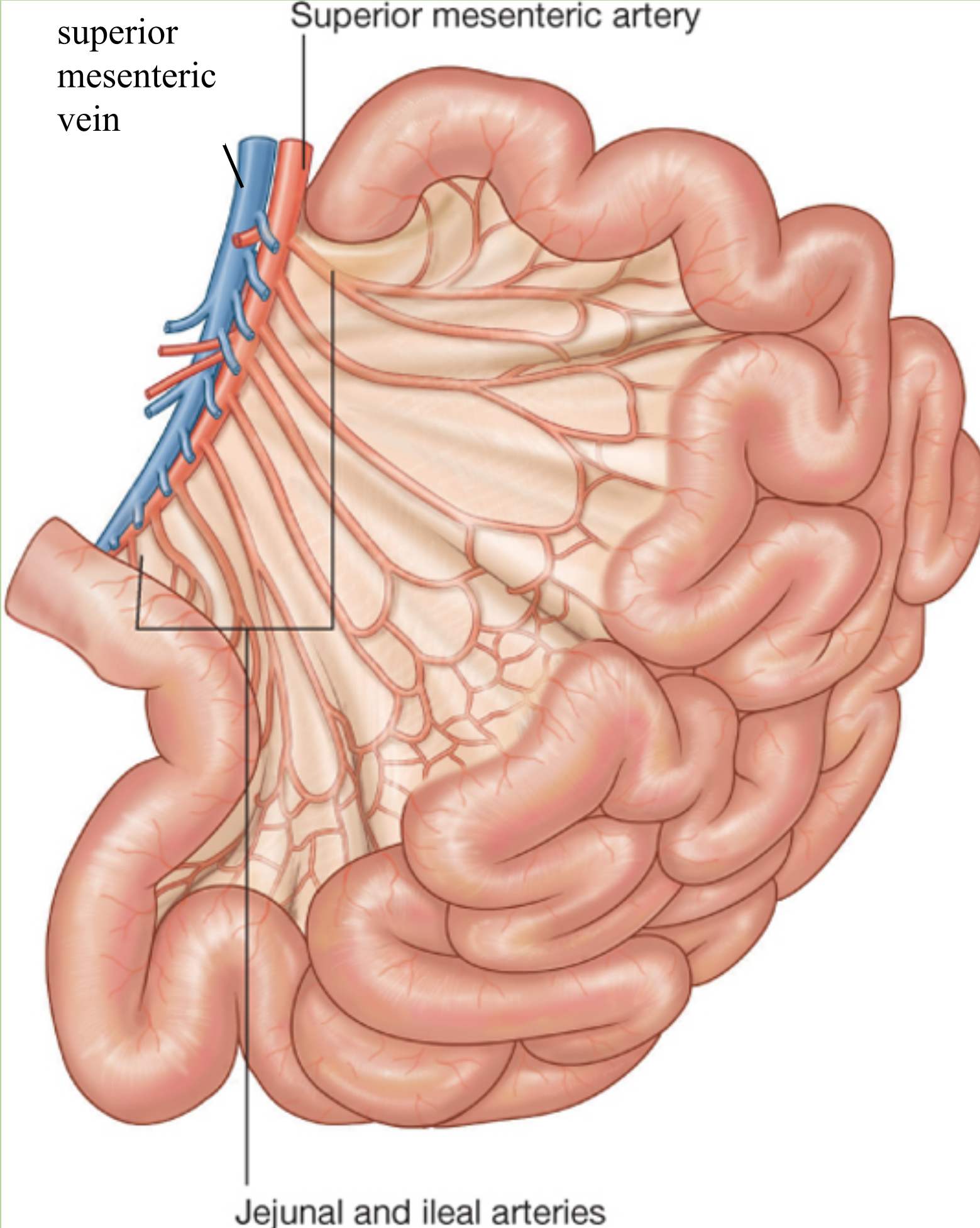
pyloric, hepatic and superior mesenteric lymph nodes (into the celiac lymph nodes)

Innervation:

parasympathetic: vagus nerve

sympathetic: splanchnic nerves





Neurovasculature of the jejunum and ileum

- jejunal arteries
- ileal arteries
- ileocolic artery

Venous drainage:

into the hepatic portal vein via superior mesenteric vein

Lymphatic drainage:

superior mesenteric lymph nodes

Innervation:

parasympathetic: vagus nerve

sympathetic: splanchnic nerves

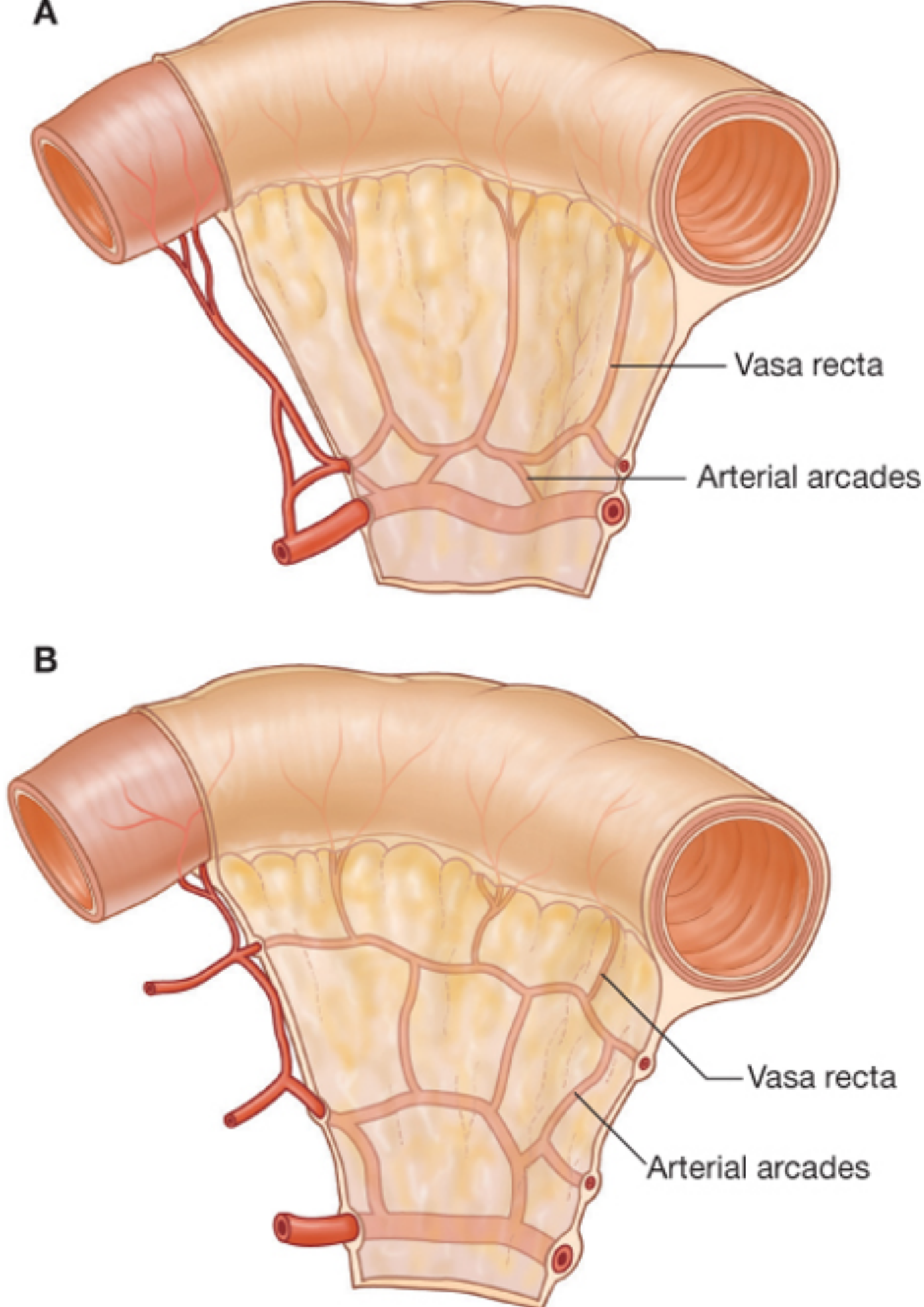
Neurovasculature of the jejunum and ileum

Jejunum:

- 1-2 series of arcades from the jejunal arteries
- the vessels enter to the intestinal wall at the **mesenteric side**

Ileum:

- 2-3 series of arcades from the ileal arteries
- the vessels enter to the intestinal wall at the **mesenteric side**



Meckel's diverticulum



- *slight bulge in the terminal ileum that results from the failure of the vitelline duct to obliterate*
- *it is considered the most common structural congenital anomaly of the gastrointestinal tract (2% of the population)*
- *usually, it is asymptomatic; most patients with a complication are thought to present in the first two years of life*

Perforated Meckel's diverticulum, Case courtesy of Dr Eric F Greif, Radiopaedia.org, rID: 34130